Ex.NO:6 INTroduction to PROLOG

Exercise 1 : Family Relationships (grandparent)

Program :

```
% Facts
parent(john, mary).
parent(mary, susan).
parent(john, mike).

% Rules
grandparent(X, Y) :- parent(X, Z), parent(Z, Y).
```

O/P

```
?- grandparent(john, susan).
true.
```

Exercise 2 : Animal Classification

Program :

```
% Facts
has_hair(tiger).
gives_milk(tiger).

% Rule
mammal(X) :- has_hair(X), gives_milk(X).
```

O/P

```
?- mammal(tiger).
true.
```

Ex. Exercise 3: Simple Arithmetic

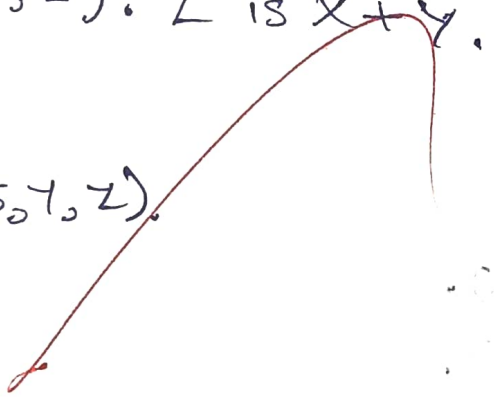
Program:

A

```
% Rule
add(X,Y,Z): Z is X+Y.
```

olp:

```
?-add(5,7,Z).
Z=12.
```



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